COLUMBIA UNIVERSITY
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STAT 5703

Statistical Inference and Modeling for Data Science

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Hypotheses Tests:
Likelihood Ratio Tests

Likelihood Ratio

Let's look closely at the likelihood ratio

$$\Lambda = \frac{L(\theta_0)}{L(\theta_a)}$$

Suppose that the true distribution of the data is $X_1, \dots, X_n \sim q(x)$, and that there are two competing models for $q(x)$, represented by the two hypotheses:

$$H_0: X_1, \dots, X_n \sim p_0(x) \; vs. \, H_a: X_1, \dots, X_n \sim p_a(x)$$

Then

$$\Lambda = \prod_{i=1}^{n} \frac{p_0(x_i)}{p_a(x_i)}$$

Define the "normalized" log likelihood ratio as

$$l_n = \frac{1}{n} \sum_{i=1}^{n} \log \frac{p_0(x_i)}{p_a(x_i)}$$

Likelihood Ratio

Now, by the LLN

$$l_n \to E(l_n) = \int \log \frac{p_0(x)}{p_a(x)} q(x)\, dx$$

$$= \int \left[\log \frac{q(x)}{p_a(x)} - \log \frac{q(x)}{p_0(x)} \right] q(x) dx$$

$$= KL(q|p_a) - KL(q|p_0)$$

where $\mathrm{KL}(q|p)$ is the KL divergence of p from q.

That is, for large n, the decision rule $\frac{L(p_0)}{L(p_a)} \leq k$ is approximately performing the comparison

$$KL(q|p_a) - KL(q|p_0) < c$$

This means we are selecting the alternative H_a when its model is "closer" to q in the sense of KL divergence!

Aside: KL divergence

One way to measure the dissimilarity between two probability distributions, p and q, is known as the *Kullback-Leibler divergence* (KL divergence) or relative entropy. This is defined as follows:

$$\mathrm{KL}(q|p) = \int q(x) \log \frac{q(x)}{p(x)} dx$$

Typically, $q(x)$ represents the "true" distribution of the data. The pdf $p(x)$ then represents a theory, model, description, or approximation of $q(x)$, and $\mathrm{KL}(q|p)$ measures how close is p to q.

Note: KL is not a proper distance, because it is not symmetric and does not satisfy the triangular inequality.

Duality between CI and hypothesis tests

- Suppose we want to test

$$H_0: \theta = \theta_0 \text{ vs. } H_a: \theta \neq \theta_0$$

  at level of significance α.

- Suppose further that we have a $(1 - \alpha)100\%$ CI for θ given by $[l(\boldsymbol{x}), u(\boldsymbol{x})]$, where $\boldsymbol{x} = (x_1, \ldots, x_n)$ is an iid sample.

- Then the rule

  "reject $H_0: \theta = \theta_0$ if $\theta_0 \notin [l(\boldsymbol{x}), u(\boldsymbol{x})]$"

  has significance level α.

Duality between CI and hypothesis tests

For example, let $X_1, \ldots, X_n \sim N(\mu, \sigma)$ where σ is *known*.

Then testing:

$$H_0: \mu = \mu_0 \; vs. \; H_a: \mu \neq \mu_0$$

at significance level α is equivalent to checking if

$$\mu_0 \in \left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$$

If $\mu_0 \notin \left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}\right]$, the reject $H_0: \mu = \mu_0$

Note: This works only for two-tail tests!

Recall the General Hypotheses Setup

Hypotheses:

$$H_0: \theta \in \Theta_0 \text{ vs. } H_a: \theta \in \Theta_a$$

where $\Theta_0 \cup \Theta_a = \Theta$ and $\Theta_0 \cap \Theta_a = \emptyset$

That is, we must decide between two regions of the sample space, which are non-overlapping and make up the the entire space.

Likelihood Ratio Test (LRT)

Definition: The likelihood ratio is defined as

$$\Lambda(\boldsymbol{x}) = \frac{L\left(\hat{\theta}_0\right)}{L\left(\hat{\theta}_{\mathrm{MLE}}\right)}$$

where $L(\theta) = f(x_1, \ldots, x_n; \theta)$ is the likelihood function, $\boldsymbol{x} = (x_1, \ldots, x_n)$ is the sample, and

$$L\left(\hat{\theta}_0\right) = \max_{\theta \in \Theta_0} L(\theta).$$

Then

$$\mathrm{RR} = \{\boldsymbol{x} \in \mathbb{R}^n : \Lambda(\boldsymbol{x}) \leq k\}$$

where k is chosen such that, the test has level α.

Note: The critical value $k < 1$, otherwise all $\boldsymbol{x}$ satisfy RR.

Example 1

Let

$$X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$$

Find the LR test for testing

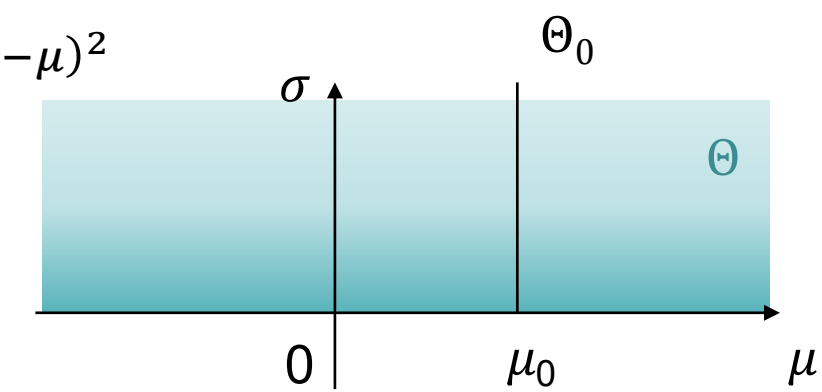
$$\mathrm{H}_0: \mu = \mu_0 \text{ vs. } \mathrm{H}_a: \mu \neq \mu_0$$

Solution:

Note that the parameter is a vector $\theta = (\mu, \sigma)$, where $\theta \in \Theta = \mathbb{R} \times \mathbb{R}_+$

$$L(\mu, \sigma) = \prod_{i=1}^{n} \left[(2\pi)^{-\frac{1}{2}} (\sigma^2)^{-\frac{1}{2}} \mathbf{e}^{-\frac{1}{2}\sigma^{-2}(x_i-\mu)^2} \right]$$

$$= (2\pi\sigma^2)^{-\frac{n}{2}} \mathbf{e}^{-\frac{1}{2\sigma}2 \sum_{i=1}^{n}(x_i-\mu)^2}$$



Example 1 (solution):

Recall: $\boxed{\hat{\mu}_{MLE} = \bar{x}, \quad \hat{\sigma}^2_{MLE} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2}$

$$L(\hat{\theta}_{MLE}) = L(\bar{x}, \hat{\sigma}^2_{MLE})$$

$$= \left[2\pi \frac{1}{n}\sum(x_i - \bar{x})^2\right]^{-\frac{n}{2}} \cdot e^{-\frac{1}{2}\cdot\frac{\cancel{\sum(x_i-\bar{x})^2}}{\frac{1}{n}\cancel{\sum(x_i-\bar{x})^2}}}$$

$$L(\hat{\theta}_0) = \max_{\sigma^2} L(\mu_0, \sigma^2)$$

Recall: MLE of σ^2 when μ is known: $\hat{\sigma}_0^2 = \frac{1}{n}\sum_{i=1}^{n}(x_i - \mu_0)^2$

Example 1 (Solution continued)

That is, for $L(\hat{\theta}_0)$ we substitute $\hat{\sigma}_0^2 = \frac{1}{n}\sum_i^n (x_i - \mu_0)^2$ and $\mu = \mu_0$ into

$$L(\mu, \sigma) = (2\pi\sigma^2)^{-\frac{n}{2}} \mathbf{e}^{-\frac{1}{2\sigma^2}\sum_{i=1}^n (x_i - \mu)^2}$$

Therefore,

$$L(\hat{\theta}_0) = \left[2\pi \frac{1}{n}\sum_{i=1}^n (x_i - \mu_0)^2\right]^{-\frac{n}{2}} \mathbf{e}^{-\frac{1}{2}\frac{\sum (x_i - \mu_0)^2}{\frac{1}{n}\Sigma (x_i - \mu_0)^2}}$$

Then the power of the exponent simplifies and:

$$= (2\pi)^{-\frac{n}{2}} n^{\frac{n}{2}} \left[\sum_{i=1}^n (x_i - \mu_0)^2\right]^{-\frac{n}{2}} \mathbf{e}^{-\frac{n}{2}}$$

Example 1 (Solution continued)

$$\Rightarrow L(\hat{\theta}_0) = (2\pi)^{-\frac{n}{2}} n^{\frac{n}{2}} \left[\sum_{i=1}^{n} (x_i - \mu_0)^2 \right]^{-\frac{n}{2}} \mathbf{e}^{-\frac{n}{2}}$$

Next, we obtain the likelihood ratio:

$$\Rightarrow \Lambda(x) = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{\mathrm{MLE}})}$$

$$= \frac{(2\pi)^{-\frac{n}{2}} n^{\frac{n}{2}} [\sum_{i=1}^{n} (x_i - \mu_0)^2]^{-\frac{n}{2}} \mathbf{e}^{-\frac{n}{2}}}{(2\pi)^{-\frac{n}{2}} n^{\frac{n}{2}} \left[\sum_{i=1}^{n} (x_i - \bar{x})^2\right]^{-\frac{n}{2}} \mathbf{e}^{-\frac{n}{2}}}$$

$$= \left[\frac{\sum (x_i - \mu_0)^2}{\sum (x_i - \bar{x})^2} \right]^{-\frac{n}{2}}$$

Example 1 (Solution):

Recall:

$$\sum (x_i - \mu_0)^2 = n(\bar{x} - \mu_0)^2 + \sum (x_i - \bar{x})^2$$

$$= n(\bar{x} - \mu_0)^2 + (n-1)s^2$$

$$\Rightarrow \Lambda(x) = \left[\frac{\sum (x_i - \mu_0)^2}{\sum (x_i - \bar{x})^2}\right]^{-\frac{n}{2}}$$

$$= \left[\frac{n(\bar{x} - \mu_0)^2 + (n-1)s^2}{(n-1)s^2}\right]^{-\frac{n}{2}}$$

$$= \left[1 + \frac{n(\bar{x} - \mu_0)^2}{(n-1)s^2}\right]^{-\frac{n}{2}}$$

$$\Rightarrow \Lambda(x) \leq k \iff 1 + \frac{n(\bar{x} - \mu_0)^2}{(n-1)s^2} \geq k_1$$

Example 1: Solution

$$\Leftrightarrow \frac{n(\bar{x}-\mu_0)^2}{(n-1)s^2} \geq k_2 = k_1 - 1$$

$$\Leftrightarrow \frac{(\bar{x}-\mu_0)^2}{s^2/n} \geq k_4 = k_3(n-1)$$

$$\Leftrightarrow \frac{|\bar{x}-\mu_0|}{s/\sqrt{n}} \geq k_5 = \sqrt{k_4}$$

$= T \sim t_{n-1}$

crit. t for two-tail α from t-table

Example 2

Let

$$X \sim \text{Binom}(n, p)$$

Find the LRT for testing

$$\text{H}_0\text{: } p = p_0 \text{ vs. } \text{H}_a\text{: } p \neq p_0$$

In this case the parameter is $\theta = p \in [0, 1]$

$$\Theta_0 = \{p_0\}, \Theta_a = [0, p_0) \cup (p_0, 1]$$

$$\hat{p}_{\text{MLE}} = \frac{x}{n} = \hat{p} = \text{sample proportion}$$

The likelihood function is

$$L(p) = \binom{n}{x} p^x (1-p)^{n-x}$$

Example 2: Solution

$$L(\hat{p}_{MLE}) = \binom{n}{x} \hat{p}^{x} (1-\hat{p})^{n-x}$$

$$L(p_0) = \binom{n}{x} p_0^{x} (1-p_0)^{n-x}$$

$$\Rightarrow \Lambda(x) = \frac{\cancel{\binom{n}{x}} p_0^{x} (1-p_0)^{n-x}}{\cancel{\binom{n}{x}} \hat{p}^{x} (1-\hat{p})^{n-x}}$$

$$= \frac{p_0^{x} (1-p_0)^{n-x}}{\hat{p}^{x} (1-\hat{p})^{n-x}} \leq K$$

Example 2: Solution

$$\Rightarrow \Lambda(x) = \frac{p_0^x (1-p_0)^{n-x}}{\left(\frac{x}{n}\right)^x \left(\frac{n-x}{n}\right)^{n-x}} \leq k$$

$$= \underbrace{\left(\frac{p_0 n}{x}\right)^x \left(\frac{n(1-p_0)}{n-x}\right)^{n-x}}_{\text{unknown distribution}} \leq k$$

Theorem (Asymptotic LRT):

Under regularity conditions, as $n \to \infty$,

$$G^2 = -2 \log \Lambda(X) \xrightarrow{d} \chi_k^2$$

where the degrees of freedom

$k = \dim(\Theta) - \dim(\Theta_0)$

Example 2 (continued)

$$-2\log\Lambda(X) = 2\log\frac{l(\hat{p})}{l(p_0)}$$

$$= 2\left\{\log\left[\left(\frac{\hat{p}}{p_0}\right)^X\right] + \log\left[\left(\frac{1-\hat{p}}{1-p_0}\right)^{n-X}\right]\right\}$$

$$= 2\left[X\log\frac{\hat{p}}{p_0} + (n-X)\log\frac{1-\hat{p}}{1-p_0}\right]$$

$$\Theta = [0,1] \subset \mathbb{R}^1 \Rightarrow \dim\Theta = 1$$

$$\Theta_0 = \{p_0\} \Rightarrow \dim\Theta_0 = 0 \Rightarrow df = 1 - 0 = 1$$

$$G^2 \to \chi_1^2$$

Example 3

Let

$$X_1, \ldots, X_n \sim \text{Exp}(\theta)$$

Find the LRT for testing

$$\text{H}_0\text{: } \theta = \theta_0 \text{ vs. H}_a\text{: } \theta > \theta_0$$

Solution: Note the that in this case we are working with a restricted parameter space

$$\Theta = [\theta_0, \infty) \text{ and } \Theta_0 = \{\theta_0\}$$

Recall the exponential pdf $f(x;\theta) = \theta e^{-\theta x}$

$$\Rightarrow L(\theta) = \prod_{i=1}^{n} (\theta e^{-\theta x_i}) = \theta^n e^{-\theta \sum_{i=1}^{n} x_i}$$

We derived earlier that $\hat{\theta}_{\text{MLE}} = \frac{1}{\bar{x}}$, when $\Theta = (0, \infty)$

Example 3 Solution

$$\hat{\theta}_{\text{MLE}} = \frac{1}{\bar{x}}\text{ , when } \Theta = (0, \infty)$$

This means that the function $L(\theta)$ has a global max at $\theta = \frac{1}{\bar{x}}$

Note L is increasing from 0 to $\frac{1}{\bar{x}}$ (you can check this formally with derivative) and $L(\theta)$ is decreasing from $\frac{1}{\bar{x}}$ to ∞.

However, we need the max in the interval $[\theta_0, \infty)$

$$\max_{[\theta_0,\infty)} L(\theta) = \begin{cases} L(1/\bar{x}) = \bar{x}^{-n}e^{-n}, \theta_0 \leq \dfrac{1}{\bar{x}} \\ L(\theta_0) = \theta_0^n e^{-n\theta_0\bar{x}}, \theta_0 > \dfrac{1}{\bar{x}} \end{cases}$$

Under the null hypothesis, $L(\theta_0) = \theta_0^n \mathbf{e}^{-n\theta_0\bar{x}}$

$$\Rightarrow \Lambda(x) = \frac{L(\theta_0)}{\max\limits_{[\theta_0,\infty)} L(\theta)} = \begin{cases} \theta_0^n \bar{x}^n e^n e^{-n\theta_0\bar{x}}, \theta_0 \leq \dfrac{1}{\bar{x}} \\ 1, \theta_0 > \dfrac{1}{\bar{x}} \end{cases}$$

Example 3 Solution

We need to solve $\Lambda(x) \leq k$ for some $k < 1$.
However,

$$\Lambda(x) \leq k \Leftrightarrow \theta_0^n \bar{x}^n \mathbf{e}^{-n\theta_0 \bar{x}} \leq k$$

Define the function g as:

$$g(\bar{x}) = \bar{x}^n \mathbf{e}^{-n\theta_0 \bar{x}}$$

Note that

$$\Rightarrow g' = n\bar{x}^{n-1}\mathbf{e}^{-n\theta_0 \bar{x}} - \bar{x}^n (n\theta_0)\mathbf{e}^{-n\theta_0 \bar{x}}$$
$$= n\bar{x}^{n-1}\mathbf{e}^{-n\theta_0 \bar{x}}(1 - \bar{x}\theta_0) > 0$$

That is, g is an increasing function. This means that

$$g(\bar{x}) \leq k \Leftrightarrow \bar{x} \leq g^{-1}(k)$$

$$\Rightarrow \mathrm{RR} = \{\bar{X} \leq c\}$$

Exercise 1

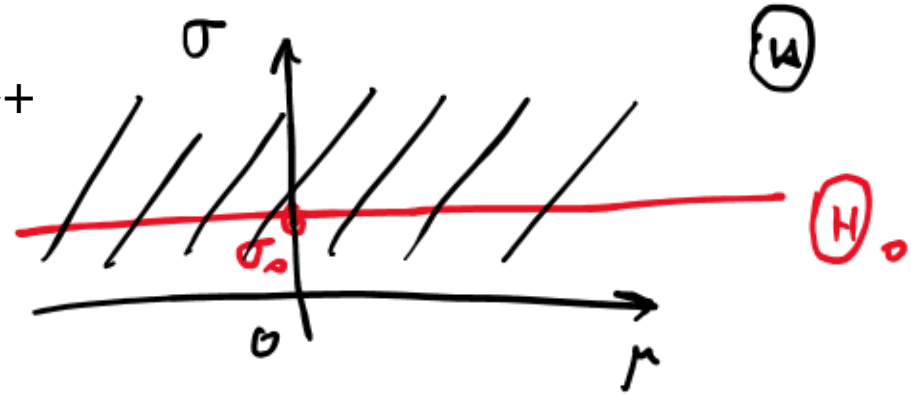
Let

$$X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$$

Find the LRT for testing

$$\mathrm{H}_0\colon \sigma^2 = \sigma^2{}_0 \text{ vs. } \mathrm{H}_a\colon \sigma^2 \neq \sigma^2{}_0$$

Note that $\Theta = \mathbb{R} \times \mathbb{R}_+$



$\dim(\Theta) = 2$

$\dim(\Theta_0) = 1$

Example 4: Two-sample *t*-test

Let $X_1, \ldots, X_m \sim \mathrm{N}(\mu_1, \sigma)$ and independently $Y_1, \ldots, Y_n \sim \mathrm{N}(\mu_2, \sigma)$. Note the two normal populations have the *same standard deviation*!

Find the LRT for testing

$$\mathrm{H}_0: \mu_1 = \mu_2 \text{ vs. } \mathrm{H}_a: \mu_1 \neq \mu_2$$

Note: Do this as exercise only if you have too much free time.
It is a difficult problem!

Example 4 (Answer):

The LRT for testing

$$H_0: \mu_1 = \mu_2 \text{ vs. } H_a: \mu_1 \neq \mu_2$$

has

Test Statistic:

$$\boxed{t = \frac{\bar{x} - \bar{y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}}$$

Pooled standard deviation:

$$s_p = \sqrt{\frac{(m-1)s_x^2 + (n-1)s_y^2}{m+n-2}}$$

Example 4 (Remarks):

The test statistic is:

$$t = \frac{\bar{x} - \bar{y}}{s_p\sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}$$

The distribution $t \sim t_{m+n-2}$ is valid under very restrictive assumptions:

- Samples each come from normal populations
- The populations have identical variances.
- Samples are independent

Q: What can you do if the assumptions are violated?

Example 4 (Assumptions):

- Samples each come from *normal* populations
- The populations have *equal variances*.
- Samples are *independent*.

Q: What can you do if the assumptions are violated?

A: For large samples (both), you can use Wald result:

The test statistic is:

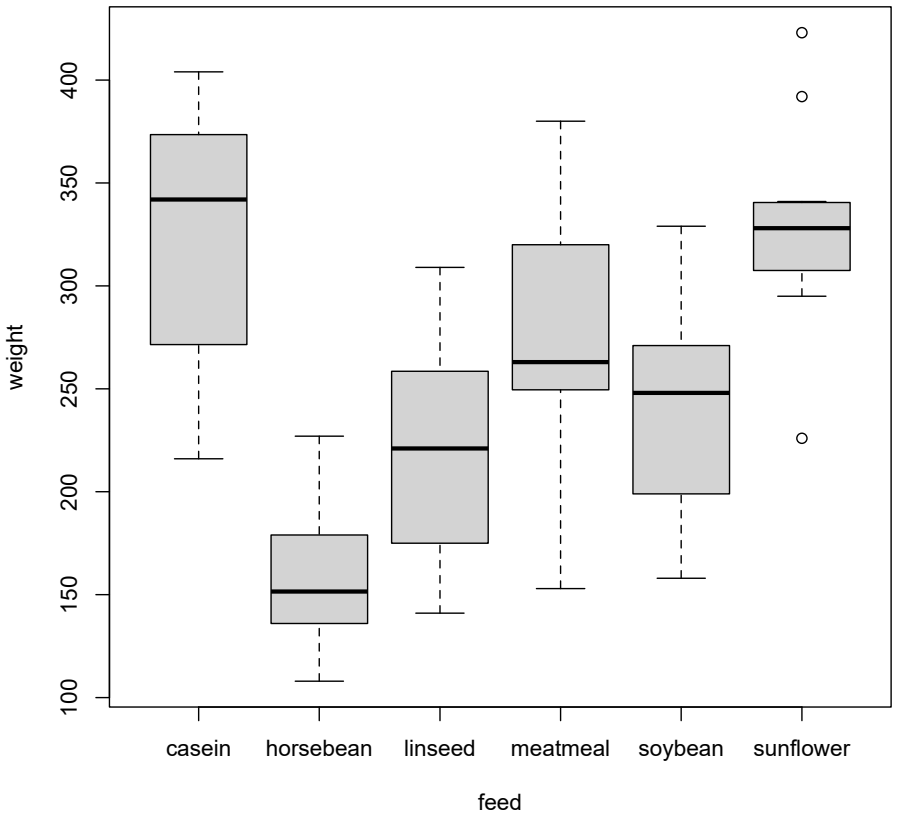
$$z = \frac{\bar{x} - \bar{y}}{\sqrt{\frac{s_x^2}{m} + \frac{s_y^2}{n}}} \approx N(0, 1)$$

Note: you still must assume independent samples.

For small samples you can use a permutation test.

Permutation Test Example

We use the `chickwts` data in R. Weights in grams are recorded, for six groups of newly hatched chicks fed different supplements for 6 weeks. There are six types of feed supplements.



The plot suggests that soybean and linseed groups may be similar.

Permutation Test Example

These are the soybean and linseed groups:

```
> x
 [1] 158 171 193 199 230 243 248 248 250 267 271 316 327 329
> y
 [1] 141 148 169 181 203 213 229 244 257 260 271 309
```

The groups can be compared in several ways. For example, sample means difference: 27.67857

or sample medians difference: 27

More generally, one can ask whether the distributions of the two variables differ and compare the groups by any statistic that measures a distance between two samples.

We will use the two-sample t-test as a metric.

Permutation Test Example

If the two samples are drawn from normal populations with equal variances, we can apply the two-sample t-test:

```
t.test(x, y, var.equal = T)

        Two Sample t-test

data:  x and y
t = 1.3208, df = 24, p-value = 0.199
alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:
 -15.57282  70.92996

sample estimates:
mean of x mean of y
 246.4286  218.7500
```

Permutation Test Example

- Assume, however, the distributions of the weights are unknown.
- The achieved significance level of the test statistic can be computed from the permutation distribution *without* requiring distributional assumptions.
- If the diets are identical, any split of the combined sample into $m = 14$ and $n = 12$ subgroups should result in similar group means.
- We will recalculate the value of the test statistic on many different permutations and evaluate how likely the observed $t = 1.32$ is.
- This way, we obtain the so-called *empirical p-value*.

Permutation Test Example

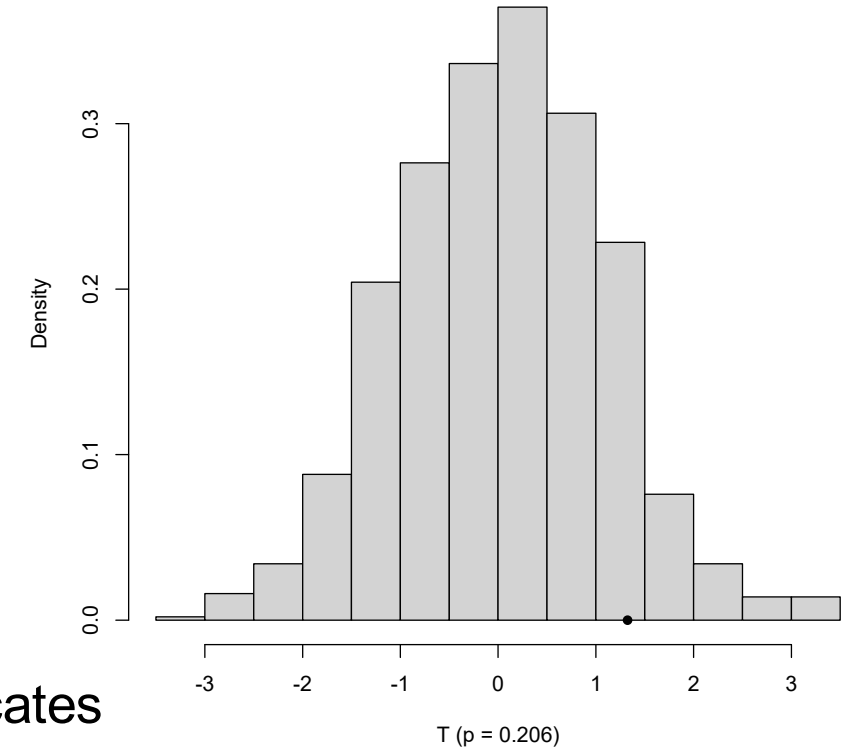
The approximate permutation test procedure is illustrated below with the two-sample *t* statistic.

```
R = 999 # number of replications of the test statistic
z = c(x, y) # vectorized data from the two samples
K <- 1:26
reps <- numeric(R); t0 <- t.test(x, y, var.equal = T)$statistic
#number of replicates
#pooled sample
#storage for replicates
for (i in 1:R) {
  #generate indices k for the first sample
  k <- sample(K, size = 14, replace = FALSE)
  x1 <- z[k]
  y1 <- z[-k] #complement of x1
  reps[i] <- t.test(x1, y1, var.equal = T)$statistic
}
p <- mean(c(t0, reps) >= t0)
> p
[1] 0.103
```

Permutation Test Example

Since we are doing a two-tail test the empirical p-value is 2*(0.103) = 0.206 so the null hypothesis is not rejected. For comparison, the two-sample t-test reports p-value = 0.198.

(The normality tests do not reject the null hypothesis so perhaps the t-test was ok)



Permutation distribution of replicates

Exercise:

A restaurant owner, wants to test if her two managers perform at the same level. To do that, she collects data about the number of customer complaints at two random samples of shifts (one for each manager).
Assume the data are normally distributed and test the hypothesis that the managers' performances are different.
Here is a summary of the results:

	Manager A	Manager B
Mean	4 complaints	5 complaints
Standard deviation	0.3	0.5
Number of shifts	19	21